App Dev Project Report

1. Student Details

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About Me: I am an executive Data Analyst working in the BFSI sector for an agency in Bahrain with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

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2. Project Details

Project Title: Placement Portal Application - V2

Problem Statement:

To build a **Placement Portal Application (PPA)** web application that allows **Admin (Institute), Companies, and Students** to interact with the system based on their roles.

Approach:

The application was developed using Flask as the backend framework and Vue.js for the frontend. It follows a modular architecture using Flask Blueprints and SQLAlchemy ORM. JWT-based authentication enables secure role-based access for Admins, Companies, and Students. Redis and Celery were integrated for asynchronous batch jobs, scheduled reminders, report generation, and caching.

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3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, creating API documentation samples, and improving variable naming consistency.
The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and documentation formatting**.
All final implementation logic, debugging, and integration were done manually.

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4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend REST APIs
SQLAlchemy	ORM
SQLite	Database
Vue.js	Frontend
Bootstrap 5	UI
Jinja2	HTML Templates
Flask-JWT-Extended	JWT Authentication
Flask-Mail	Email Services
Flask-Caching	Redis Caching
Redis	Cache & Celery Broker
Celery	Background Jobs
Celery Beat	Scheduled Tasks

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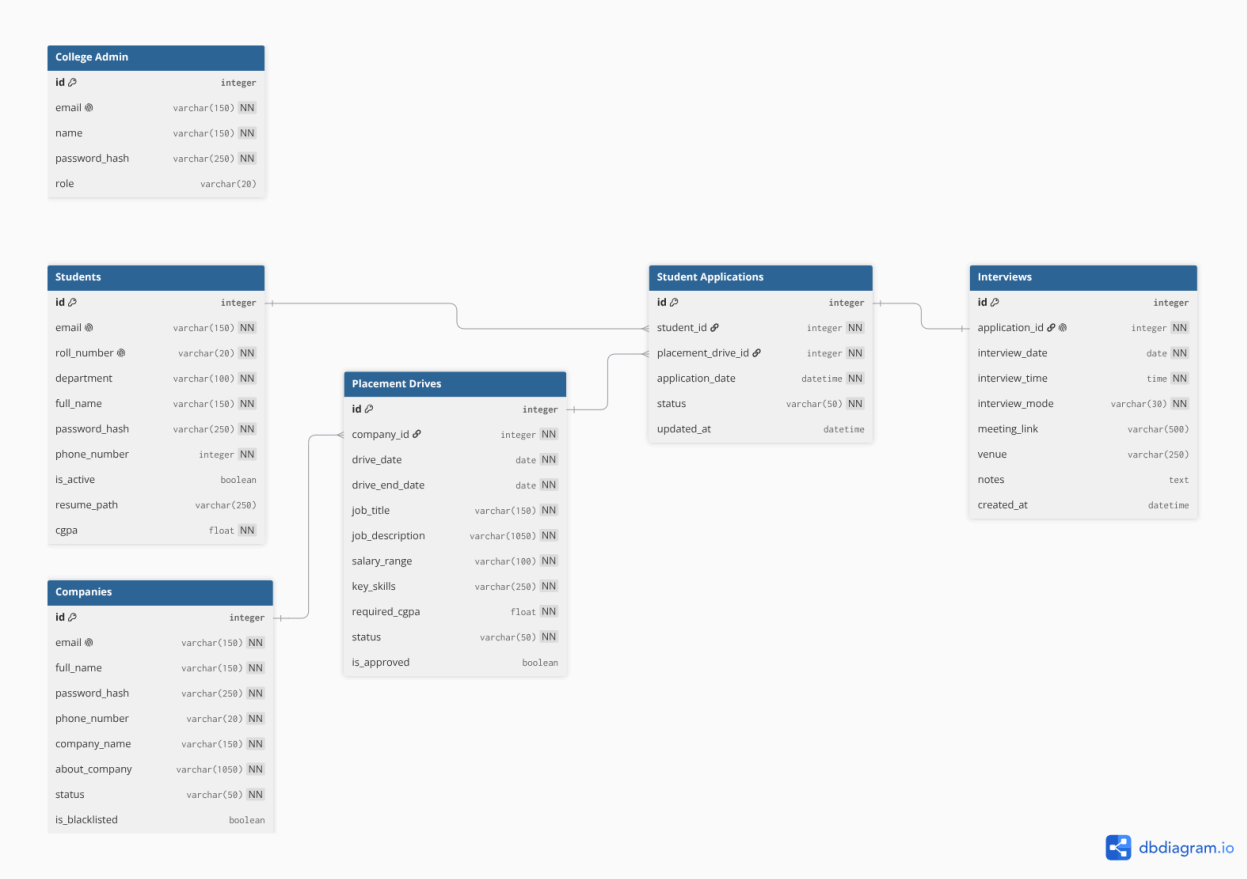
5. Database Schema / ER Diagram

Tables:

1. **College Admins**— stores user details of an admin (id, email, name, passhash)
2. **Students** — stores user details of a student (id, email, roll number, department, full name, password hash, phone number, activity status, resume path, cgpa)
3. **Companies** — stores user details of a company (id, email, full name, password hash, phone number, company name, about company, status, is blacklisted)
4. **Placement Drives** — stores the details of placement drives created by companies (id, company id, drive date, drive end date, job title, job description, salary range, key skills, required CGPA, status, is approved)
5. **Student Applications**— stores application details submitted by a student to a placement drive (id, student id, placement drive id, application date, status, updated at)
6. **Interviews** — stores interview details of a placement drive and a student (id, application id, interview date, interview time, interview mode, meeting link, venue, notes, created at)

Relationships:

- One-to-Many → Companies → Placement Drives
- One-to-Many → Students → Applications
- One-to-One → Applications → Interviews
- Many-to-One → Applications → Placement Drives



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6. API Resource Endpoints

1. Authentication

Method	Endpoint	Purpose
POST	/api/auth/login/<role>	User login
POST	/api/auth/signup/student	Student registration
POST	/api/auth/signup/company	Company registration

2. **Student Module**

Method	Endpoint	Purpose
GET/POST	/api/student/dashboard	Student dashboard
GET/POST	/api/student/view_drive/<id>	View placement drive
POST	/api/student/apply_to_drive/<id>	Apply for a drive
GET/POST/PUT	/api/student/edit_profile/<id>	Edit profile
POST	/api/student/export_csv	Export application history
GET	/api/student/export_status/<task_id>	Check export status

3. **Company Module**

Method	Endpoint	Purpose
GET/POST	/api/company/dashboard	Company dashboard
GET/POST	/api/company/create_drive	Create placement drive
GET/POST	/api/company/view_drive/<id>	View drive and applicants
PUT	/api/company/change_application_status/<id>	Update application status
GET/POST/PUT	/api/company/edit_profile/<id>	Edit company profile

4. **Admin Module**

Method	Endpoint	Purpose
GET/POST	/api/admin/dashboard	Admin dashboard
PUT	/api/admin/change_company_status/<id>	Approve/Reject company
PUT	/api/admin/change_drive_status/<id>	Approve/Reject drive
GET/POST	/api/admin/student_profile/<id>	View student profile
GET/POST	/api/admin/student_applications/<id>	View student's applications

5. **Common Path**

Method	Endpoint	Purpose
POST	/api/auth/search	Search students, companies, drives and applications

GET/POST	/api/*/view_drive/<id>	Shared placement drive view

7. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **model.py** – database models using SQLAlchemy
- **routes.py** – Flask Blueprints for user and activity routes
- **frontend/static/** – Javascript scripts
- **frontend/templates/ -** HTML Templates

Implemented Features:

- User registration (for students and companies) and login.
- Placement drive registration after approval from admin.
- Student profile and resume storage.
- Student applications on eligible placement drives.
- Interview scheduling by companies after students' profile/resume review.
- Scheduled notifications to inform students of interviews.
- Scheduled monthly report generation for admin.
- Responsive UI built with Bootstrap.
- Secure password hashing and user session handling.

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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1K1WaS0DTsBx6in1aeNaTZbZcM5NmM657/view?usp=sharing

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